Additional Experimental Results in Authors' Response

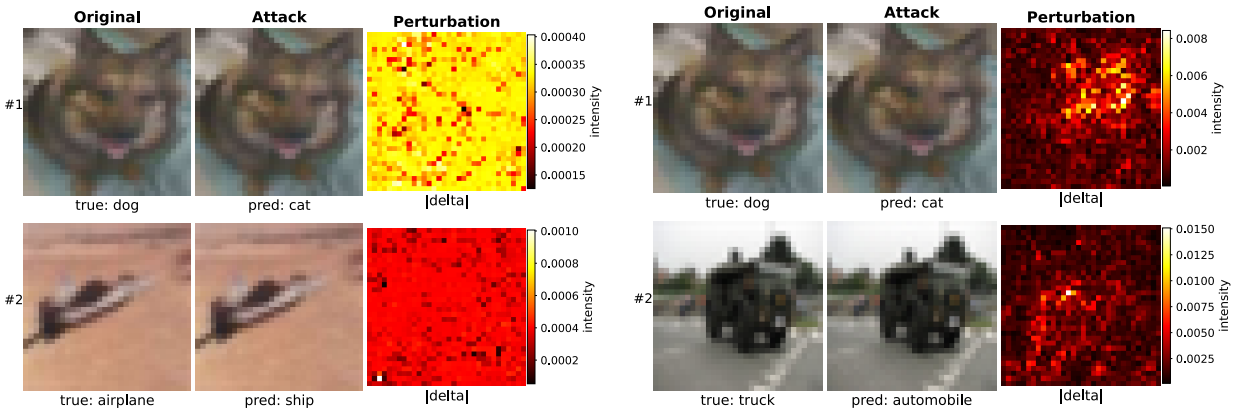
1 ML Application: Non-Uniform Adversarial Attack

We evaluate our method on non-uniform adversarial attacks with non-convex perturbation regions [1]. The PGD baseline used radial feasibility scaling (non-euclidean projection) rather than exact orthogonal projection [1]; our Hom-PGD$^+$ reparameterizes the perturbation region to the unit ball.

Setup. CIFAR-10 with a trained CNN classification model. For each clean input $x \in [0,1]^{64\times64\times3}$, perturbations are constrained by $\Delta(x) = \left\{\delta \in \mathbb{R}^d : \|W\delta\|_p \le \epsilon_p,\ 0 \le x+\delta \le 1\right\}$, where W is a positive-definite weighting operator (here: channel-spatial structure combining inter-channel correlation and spatial correlation), and ϵ_p is the preset perturbation level. The adversarial attack is performed over $N = 500$ correctly-classified samples with step-size $\alpha = 10^{-3}$ and 500 iterations.

Table 1: Attack performance under weighted ℓ_p perturbation constraints on CIFAR-10.

p	Method	Adv. Acc. (%) ↓	Success (%) ↑	Time/iter (s)
0.5	PGD	98.0	2.0	0.042
	Hom-PGD$^+$	**85.4**	**14.6**	0.042
1	PGD	97.2	2.8	0.041
	Hom-PGD$^+$	**83.6**	**16.4**	0.042
2	PGD	79.4	20.6	0.041
	Hom-PGD$^+$	80.0	20.0	0.042
4	PGD	78.6	21.4	0.041
	Hom-PGD$^+$	79.2	20.8	0.042



(a) PGD (b) Hom-PGD

Figure 1: Successful adversarial examples under weighted $\ell_{1/2}$ perturbation constraints.

Key finding. Hom-PGD$^+$ achieves 7× higher success rate for non-convex regions ($p \le 1$) with negligible overhead, while performing comparably for convex regions ($p \ge 2$). PGD struggles in the non-convex setting because projection, Euclidean or otherwise, onto a non-convex region may lead to convergence issues [2], whereas our methods transform the feasible region into a simple unit ball with better constraint geometry.

2 Additional Baselines for the Chance Constrained Problems

We add two convex approximation for the chance constraint $\mathbb{P}[g(x,\xi) \le 0] \ge 1-\epsilon$ with scenarios $\{\xi^{(i)}\}_{i=1}^N$:

- Robust Scenario: $g(x,\xi^{(i)}) \le 0,\ \forall\, i \in \{1,\dots,N\}$.
- CVaR Approximation: $t + \frac{1}{\epsilon N}\sum_{i=1}^N [g(x,\xi^{(i)}) - t]^+ \le 0$, where $[\cdot]^+ = \max\{\cdot, 0\}$.

Given the convex nature of both formulations, we use MOSEK with an academic license to solve them.

Table 2: JCC-OPF baseline results (200-bus system, 100 scenarios).

Solver	Status	Time (s)	Objective	Feas. Rate
CVaR	optimal	73.7	0.683	1.00
Robust	optimal	74.0	0.684	1.00

Scalability limitation. Our experiments reveal that even these convex formulations do not scale well when the number of scenarios increases. On the 200-bus system with 100 scenarios, CVaR requires 73.7 seconds and the robust formulation requires 74.0 seconds. However, for 1,000 scenarios or larger bus systems, MOSEK returns a failure flag. This indicates that direct application of these convex relaxations becomes infeasible at practical scales, motivating the need for scalable iterative methods such as ours.

3 Extended Scaling Experiments on QCQP

We extend our scaling analysis with a larger 100-dimensional QCQP instance. The problem complexity scales as $O(n_{\text{qua}} \cdot n_{\text{var}}^2)$, where n_{qua} denotes the number of quadratic constraints and n_{var} the number of variables.

Table 3: QCQP scaling experiments ($n_{\text{var}} = 100$).

n_{qua}	Hom-PGD$^+$ Obj	IPOPT Obj	Hom-PGD$^+$ (s)	IPOPT (s)	Speedup
10	-0.470	-0.483	12.0	2.3	$0.2\times$
100	-0.455	-0.499	12.1	25.8	$\mathbf{2.1}\times$
1000	-0.453	-0.430	12.3	255.6	$\mathbf{20.7}\times$

Key Finding. Hom-PGD$^+$ computation time remains nearly constant as the constraint count grows, achieving $>20\times$ speedup at 1000 constraints while maintaining comparable objective quality. Combined with the JCC-OPF results in our original manuscript (e.g., $12\times$ speedup on 500-bus systems), this confirms the order-of-magnitude improvements at scale claimed in our abstract.

References

[1] Ecenaz Erdemir, Jeffrey Bickford, Luca Melis, and Sergul Aydore. Adversarial robustness with non-uniform perturbations. *Advances in Neural Information Processing Systems*, 34:19147–19159, 2021.

[2] Rina Foygel Barber and Wooseok Ha. Gradient descent with non-convex constraints: local concavity determines convergence. *Information and Inference: A Journal of the IMA*, 7(4):755–806, 2018.